

Flavivirus RNA in phlebotomine sandflies.

PMID: 19492949 | Indexed in PubMed

Sandfly-transmitted phleboviruses, such as Toscana, sandfly fever Sicilian, and sandfly fever Naples, can cause human disease and circulate at high rates in Mediterranean countries. Previous studies have also established that viruses other than phleboviruses may be detected in and isolated from sand flies. The recent detection and isolation (in a large variety of mosquito species) of insect-only flaviviruses related to cell fusing agent virus has indicated that the latter is not an evolutionary remnant but the first discovered member of a group of viruses, larger than initially assumed, that has high genetic heterogeneity. Insect-only flaviviruses have been detected in and/or isolated from various species of mosquitoes, but nevertheless only from mosquitoes to date; other dipterans have not been screened for the presence of insect-only flaviviruses. The possible presence of flaviviruses, including insect-only flaviviruses, was investigated in sand flies collected around the Mediterranean during a trapping campaign already underway. Accordingly, a total of 1508 sand flies trapped in France and Algeria, between August 2006 and July 2007, were tested for the presence of flaviviruses using a PCR assay previously demonstrated experimentally to amplify all recognized members of the genus *Flavivirus*, including insect-only flaviviruses. Two of 67 pools consisting of male *Phlebotomus perniciosus* trapped in Algeria were positive. The two resulting sequences formed a monophyletic group and appeared more closely related to insect-only flaviviruses associated with *Culex* mosquitoes than with *Aedes* mosquitoes, and more closely related to insect-only flaviviruses than to arthropod-borne or to no-known-vector vertebrate flaviviruses. This is the first description of insect-only flaviviruses in dipterans distinct from those belonging to the family Culicidae (including *Aedes*, *Culex*, *Mansonia*, *Culiseta*, and *Anopheles* mosquito genera), namely sand flies within the family Psychodidae. Accordingly, we propose their designation as phlebotomine-associated flaviviruses.

Acetic acid bacteria, newly emerging symbionts of insects.

PMID: 20851977 | Indexed in PubMed

Recent research in microbe-insect symbiosis has shown that acetic acid bacteria (AAB) establish symbiotic relationships with several insects of the orders Diptera, Hymenoptera, Hemiptera, and Homoptera, all relying on sugar-based diets, such as nectars, fruit sugars, or phloem sap. To date, the fruit flies *Drosophila melanogaster* and *Bactrocera oleae*, mosquitoes of the genera *Anopheles* and *Aedes*, the honey bee *Apis mellifera*, the leafhopper *Scaphoideus titanus*, and the mealybug *Saccharicoccus sacchari* have been found to be associated with the bacterial genera *Acetobacter*, *Gluconacetobacter*, *Gluconobacter*, *Asaia*, and *Saccharibacter* and the novel genus *Commensalibacter*. AAB establish symbiotic associations with the insect midgut, a niche characterized by the availability of diet-derived carbohydrates and oxygen and by an acidic pH, selective factors that support AAB growth. AAB have been shown to actively colonize different insect tissues and organs, such as the epithelia of male and female reproductive organs, the Malpighian tubules, and the salivary glands. This complex topology of the symbiosis indicates that AAB possess the keys for passing through body barriers, allowing them to migrate to different organs of the host. Recently, AAB involvement in the regulation of innate immune system homeostasis of *Drosophila* has been shown, indicating a functional role in host survival. All of these lines of evidence indicate that AAB can play different roles in insect biology, not being restricted to the feeding habit of the host. The close association of AAB and their insect hosts has been confirmed by the demonstration of multiple modes of transmission between individuals and to their progeny that include vertical and horizontal transmission routes, comprising a venereal one. Taken

together, the data indicate that AAB represent novel secondary symbionts of insects.

The C-terminal extension that characterizes mosquito (Diptera: Culicidae) ribosomal protein S6 is widespread among the Culicomorpha.

PMID: 17695015 | Indexed in PubMed

Eukaryotic ribosomal proteins, which participate in the structure and function of the translational machinery, are generally well conserved. In *Aedes* and *Anopheles* mosquitoes (Diptera: Culicidae), however, ribosomal protein S6 (RpS6) contains a low-complexity, C-terminal extension that is absent from its homolog in *Drosophila melanogaster* Meigen and *Manduca sexta* (L.). To explore the distribution of RpS6 C-terminal extensions in genera phylogenetically closer to mosquitoes than is *Drosophila*, we recovered cDNAs encoding RpS6 from a phantom midge (*Chaoborus* sp.), a midge (*Chironomus* sp.), a blackfly (*Simulium* sp.), and a phantom crane fly [*Bittacomorpha clavipes* (F.)]. Sequences of deduced translation products showed that RpS6 extensions occurred only in members of the Culicomorpha, and they were absent from *B. clavipes*, a member of the Ptychopteromorpha, which forms the sister group to the Culicomorpha. Likewise, the extension did not occur in *Telamioscopus* sp., a moth fly classified among the more distantly related Psychodomorpha. The C-terminal extensions on RpS6 ranged in length from 81 to 190 amino acids, they were highly enriched for lysine and alanine, and they seem to be evolving more rapidly than the conventional portion of the RpS6 protein shared by all eukaryotes. Although analysis of RpS6 protein was consistent with analyses based on ribosomal DNA, suggesting that Chironomidae is the sister group to the remaining families in the Culicomorpha, trees generated from RpS6 amino acid sequences were largely congruent with accepted phylogenies based on morphological characters. Our results suggest that a C-terminal, lysine-rich extension on RpS6 is a potential molecular synapomorphy for the Culicomorpha.

An Inexpensive System to Investigate the Daily Rhythms of Insects.

PMID: 37796732 | Indexed in PubMed

Insects' daily rhythms occur in response to their surrounding environment. Recognizing the daily rhythms of pathogen vectors can be helpful in developing effective, safe, and sustainable management strategies to control vector insects and reduce the spread of pathogens. However, studying the daily rhythm of insects often requires costly or labor-intensive trapping, and few tools are available to quantify daily rhythms in the field. We developed a simple collection system to study the flight activity of mosquitoes and biting midges using a contained, programmable, rotating, automatic pet feeder. A diverse assemblage of nuisance and vector species were collected with our system, including mosquitoes of the genera *Aedes*, *Anopheles*, *Culex*, and *Deinocerites* and biting midges (Ceratopogonidae) such as the coastal pest *Culicoides furens*. Surprisingly, mosquitoes and biting midges were less active during crepuscular periods (1800-2100h; 0600-0900h) than during dark periods (2100h-2400h; 0300h-0600h). A number of urban and agricultural pest insects were captured, including Coleoptera, Hymenoptera, Isoptera and Lepidoptera. This study shows that relatively inexpensive products can be adapted to study the daily rhythms of flying vectors and nuisance arthropods, with implications for vector-borne disease transmission and control. The collection system could also be used with flight intercept or pitfall traps, permitting study of the circadian activity patterns of a diverse array of arthropods.

Arthropod vectors in the evolution of bunyaviruses.

PMID: 1683124 | Indexed in PubMed

Viruses from each genus of Bunyaviridae have preferential relationships to the arthropods of only one or two families, i.e. Bunyavirus to mosquitoes (Culicidae), Phlebovirus to sand flies (Psychodidae) and mosquitoes, Uukuvirus and Nairovirus to ticks (Ixodidae and Argasidae). An exception is genus Hantavirus not proven to be transmitted by vectors. Within the Bunyavirus genus 16 serogroups have been recognized on the basis of their antigenic relationship. Based on isolations from the nature each serogroup is preferentially linked with arthropod species (mostly mosquitoes) of one, or two genera. For 8 out of 16 serogroups Culex mosquitoes are the main insect vectors. Two serogroups are linked with Aedes mosquitoes, three with Anopheles mosquitoes. Aedeomyia mosquitoes, Culicoides biting midges and Hyalomma ticks are vectors of one serogroup each. Evolutionary trends within the genus Bunyavirus and within the family Bunyaviridae can be recognized based on relationships of bunyaviruses to their arthropod vectors.

Increased demand of urine cultures from Danish general practice: a five-year register-based study.

PMID: 37052881 | Indexed in PubMed

To characterise and explore the development in the number and content of urine samples sent from general practice in the North Denmark Region to the Department of Clinical Microbiology (DCM) at Aalborg University Hospital during a five-year period. A register-based study. General practice. Urine samples received at DCM, Aalborg University Hospital from general practice between 2017 and 2022. Number and content of urine samples. A total of 255,271 urine samples from general practice were received at DCM, with 76.1% being from female patients. Uropathogens were identified in 43.0% of the samples. During the five-year period, a 23.0% increase in the number of urine samples per person (incidence rate ratio (IRR) 1.23, 95% CI 1.21-1.25) was observed. A slight increase in the proportion of positive cultures (risk ratio (RR) 1.03, 95% CI 1.01-1.05) was seen. No notable change in the patient population (age, gender) was observed. Overall, Escherichia coli was the most identified uropathogen (60.4%) followed by Klebsiella spp. (8.7%) and Enterococcus spp. (7.7%). Distribution of the various uropathogens differed slightly depending on patient gender and age, importantly E. coli was less frequently observed in males aged >65 years. During the past five years an increasing amount of urine cultures have been requested at DCM from general practice. Importantly, the cause(s) of this increasing demand needs to be explored further in future studies.

Uropathogenic E. coli and Hybrid Pathotypes in Mexican Women with Urinary Tract Infections: A Comprehensive Molecular and Phenotypic Overview.

PMID: 38921024 | Indexed in PubMed

Uropathogenic Escherichia coli (UPEC) is the main cause of urinary tract infections (UTIs) and carries virulence and resistance factors often found in mobilizable genetic elements, such as plasmids or

pathogenicity islands (PAIs). UPEC is part of the extraintestinal pathogenic *E. coli* (ExPEC), but hybrid strains possessing both diarrheagenic *E. coli* (DEC) and ExPEC traits, termed "hypervirulent", present a significant health threat. This study assessed the prevalence of UPEC PAIs, ExPEC sequence types (ST), DEC genes, carbapenemase and extended-spectrum β -lactamase (ESBL) phenotypes, resistance genotypes, and plasmids in 40 clinical isolates of UPEC. Results showed that 72.5% of isolates had PAIs, mainly PAI IV536 (53%). ESBL phenotypes were found in 65% of β -lactam-resistant isolates, with 100% of carbapenem-resistant isolates producing carbapenemase. The predominant ESBL gene was blaCTX-M-2 (60%), and the most common resistance gene in fluoroquinolone and aminoglycoside-resistant isolates was aac(6')Ib (93%). Plasmids were present in 57% of isolates, and 70% belonged to the ST131 clonal group. Molecular markers for DEC pathotypes were detected in 20 isolates, with 60% classified as hybrid pathotypes. These findings indicate significant pathogenic potential and the presence of hybrid pathotypes in *E. coli* UTI clinical isolates in the Mexican population.

The etiology of urinary tract infection: traditional and emerging pathogens.

PMID: 12601338 | Indexed in PubMed

The microbial etiology of urinary infections has been regarded as well established and reasonably consistent. *Escherichia coli* remains the predominant uropathogen (80%) isolated in acute community-acquired uncomplicated infections, followed by *Staphylococcus saprophyticus* (10% to 15%). *Klebsiella*, *Enterobacter*, and *Proteus* species, and enterococci infrequently cause uncomplicated cystitis and pyelonephritis. The pathogens traditionally associated with UTI are changing many of their features, particularly because of antimicrobial resistance. The etiology of UTI is also affected by underlying host factors that complicate UTI, such as age, diabetes, spinal cord injury, or catheterization. Consequently, complicated UTI has a more diverse etiology than uncomplicated UTI, and organisms that rarely cause disease in healthy patients can cause significant disease in hosts with anatomic, metabolic, or immunologic underlying disease. The majority of community-acquired symptomatic UTIs in elderly women are caused by *E. coli*. However, gram-positive organisms are common, and polymicrobial infections account for up to 1 in 3 infections in the elderly. In comparison, the most common organisms isolated in children with uncomplicated UTI are Enterobacteriaceae. Etiologic pathogens associated with UTI among patients with diabetes include *Klebsiella* spp., Group B streptococci, and *Enterococcus* spp., as well as *E. coli*. Patients with spinal cord injuries commonly have *E. coli* infections. Other common uropathogens include *Pseudomonas* and *Proteus mirabilis*. Recent advances in molecular biology may facilitate the identification of new etiologic agents for UTI. The need for accurate and updated population surveillance data is apparent, particularly in light of concerns regarding antimicrobial resistance. This information will directly affect selection of empiric therapy for UTI.

Most frequent causes of urinary tract infections in children.

PMID: 20514772 | Indexed in PubMed

to determine the most frequent causes of urinary tract infection (UTI) amongst children according to gender and age. 299 children with urinary tract infections were evaluated. All of the cases were clinically evaluated and quantitative and qualitative examinations of the urine (culture with antibiogram) were done. There were 54.58% males and 45.15% females in the research group. Similar distribution was present in age groups: 1-28 days (67.86% male, 32.14% female), 1 month up to 1 year (64.56% male,

35.44% female), 2-6 years (55.29% male, 44.71% female). The only difference was found in the age group of 7-16 years where there were more female patients (56.07% female, 43.93% male). Based on microbiological findings, in 91.30% of the cases only one species was isolated, and more than one species was isolated in 8.70% of cases. *Escherichia coli* was the most common cause of urinary tract infection (62.88%), the second most common was *Klebsiella* (23.080%), followed by *Proteus mirabilis* (7.02%), *Citrobacter*(5.35%), *Staphylococcus saprophyticus* (1.34%) and *Candida albicans* (0.33%). *E. coli* was the most frequent causative of UTI amongst children of all ages. In the age group of newborns, *E. coli* was isolated as the etiological causative of UTI in 57.14% of cases. In the age group of month-1 year, *E.coli* was the most frequent cause of urinary tract infection (60.76%). In the age group of 2-6 years, *E.coli* was the most frequent cause of urinary tract infection (74.12%). In the age group of 7-16 years, *E. coli* was isolated in 59.81% of cases. *Klebsiella* was the second most frequent causative (23.08%) and with a higher incidence in the age group of newborns (42.86%). UTI was generally more present in male patients. *E. coli* was the most frequent cause of UTI in children of all age, while the second most frequent cause of UTI in children was *Klebsiella*.

[Urinary infection in adolescents].

PMID: 14676879 | Indexed in PubMed

OBJECTIVE: Urinary tract infections are caused by various gram-negative bacteria. *Escherichia coli* is the most common causative agent in all groups of patients. *Staphylococcus saprophyticus* is now recognized as a common cause of urinary tract infection in adolescents and young adult women. **METHODS:** Review of literature and clinical experience with urinary tract infections in adolescents at the Pediatric Nephrology Department of Hospital Santa Casa de São Paulo. **RESULTS:** Lower urinary tract infection, hematuria and dysuria were the most frequent symptoms observed in infections caused by *Staphylococcus saprophyticus*. Some asymptomatic cases were observed. The infection may be associated with adverse effects of maternal and fetal health during pregnancy. **CONCLUSIONS:** This article reviews the diagnosis, etiology, treatment, and complications associated with urinary tract infections in adolescents.
